

JAYANT TOMAR

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EDUCATION

Delhi Technological University (DTU), New Delhi, India

CGPA: 9.34

Bachelor of Technology in Computer Science and Engineering

2024 – 2028

Class 12th CBSE Board

Percentage: 95%

Class 10th CBSE Board

Percentage: 97.2%

PROJECTS

Fast Mini Search Engine | *C++, BM25, Inverted Index, Trie, Multithreading*

[GitHub](#) | [Video](#)

- Built a **multithreaded C++ search engine** using an inverted index with **BM25** ranking.
- Implemented **Trie-based autocomplete** and **Levenshtein spell correction** to improve search accuracy.
- Designed a C++ HTTP backend exposing **REST APIs** for search, snippets, autocomplete, **document upload**, indexing, search and indexing **latency** and **benchmarking** operations.
- Developed **Hybrid multithreaded indexing pipeline** with incremental **runtime ingestion** and temporary **disk-backed runtime corpus**.
- Implemented **parallel document indexing** using **thread-local inverted indexes** with **post-merge** aggregation, achieving **2.14×** indexing speedup and sub-millisecond average search latency (**0.83 ms**, **1200 queries/sec** throughput).

High-Performance In-Memory Key-Value Store | *C++, TCP*

[GitHub](#) | [Video](#)

- Engineered **thread-safe** operations using **mutex-based** synchronization, ensuring **data integrity** for over 500 concurrent clients with sub-millisecond latency (**0.305 ms** avg, $p99 < 0.9ms$).
- Architected **hybrid persistence (WAL + snapshot/RDB)** enabling crash recovery and fast restart by combining log replay with periodic full-state snapshots.
- Optimized **LRU cache eviction with heap-optimized TTL expiration ($O(\log n)$)**, replacing full scans, sustaining 55K TTL writes/sec with only **0.018 ms** avg latency..
- Developed **master-slave replication with initial synchronization**, where slaves receive snapshot-based state transfer followed by real-time command replication for consistency.
- Achieved **93K+** GET ops/sec and **60K+** SET ops/sec while integrating graceful shutdown and structured logging, and a zero-dependency **WebSocket-to-TCP proxy bridge** for real-time visual telemetry and cluster state exploration.

ACHIEVEMENTS AND AWARDS

- **Top 12 Finalist:** Hack-LLM (24-Hour Hackathon), IIIT Delhi - [Certificate](#)
- **Competitive Programming:** **Knight** (1885) at LeetCode, **Pupil** at Codeforces and **3-Star** at CodeChef.
- **Algorithmic Consistency:** Solved **600+** **DSA problems** on LeetCode, maintaining a rigorous 200-day streak.
- **Goldman Sachs India Hackathon 2026-CS:** Secured All India Rank **1127** - [Leaderboard](#)
- **JEE Advanced & Mains 2024:** Secured AIR 10,987 in JEE Advanced and **99.27** Percentile in JEE Mains (placed in the top 0.73% of over 1.5 million candidates).

TECHNICAL SKILLS

Languages: C, C++, Python, SQL, HTML, CSS

CS Fundamentals: Data Structure, Algorithms, OOPs, Operating System, DBMS, Computer Networks

Developer Tools: Git, Google Colab, Kaggle, VS Code, Visual Studio, MySQL, JupyterLab

LEADERSHIP & CO-CURRICULARS

UGV-DTU (Unmanned Ground Vehicle Team)

New Delhi, India

Software Division Member

Sept 2024 - May 2025

- Collaborated with team members on software components for robotics systems and Contributed to testing and debugging

CERTIFICATIONS

Stanford University Algorithms Specialisation

[Certificate](#)

Machine Learning Specialization by Andrew Ng

[Certificate](#)

Kaggle Intermediate ML

[Certificate](#)

Kaggle Intro to ML

[Certificate](#)